

Take-Home Task: Person-Vehicle Interaction

Task Description

You are given several short MP4 clips (no audio). Your job is to write Python code that will take a clip as input and output a list of the human-vehicle interactions in the clip. For instance, a person exiting or entering the vehicle is considered interacting with the vehicle, while a person passing by the vehicle is not. The output should be structured and machine-readable, you are free to define the format.

If an ambiguity arises, resolve it and document the decision taken, together with the alternatives considered.

You may assume that all clips are independent of each other and there is no need to maintain inter-clip identities for persons or for vehicles.

Data

- Folder: Videos

Requirements

The code should produce a machine-readable artifact that lists all person-vehicle interactions in the given set of clips including the following fields for each interaction:

- clip_id
- time span(s) or frame range(s)
- a description of the person(s) involved
- a description of the vehicle involved

Deliverables

Provide a public repository link in GitHub that contains the following:

1. Source code in Python that reproduces your results.
 - Include clear run instructions (README): environment setup, commands, and any external assets you rely on.
2. Outputs (in the repository or uploaded into a separate shared folder).
 - List of human-vehicle interactions in your chosen machine-readable format.

- Any optional visualizations you deem helpful (e.g., annotated frames) are welcome but not required.

3. Brief write-up (≤ 2 pages).

- Your approach, assumptions, limitations, and anything you would have done next given more time.

Rules & Guidelines

- Any libraries or pre-trained models may be used. If you rely on external services/APIs, document clearly.

- Optimize for clarity, determinism, and reproducibility. If randomness is involved, fix a seed.

- Keep the scope reasonable; this is a take-home, not a production system.

- The goal is not a perfect score, but rather to showcase your knowledge and problem-solving approaches. That said, a good score is certainly more welcome than a poor score :)